Đào Ngọc Hoa

daongochoa2002 | Hoa Dao | daongochoa2002.github.io |
daongochoa2002@gmail.com | +84 977 9080 27

Profile

Motivated researcher/ software engineer with a strong background in Temporal Knowledge Graphs, hybrid RAG systems, and graph-based reasoning. Experienced in building scalable AI retrieval applications and production backend systems.

Skills Summary

Programming Language	Python, Kotlin, Java, C++, JavaScript
AI & Data Engineering	PyTorch, Milvus (Vector Search), Neo4j (Graph), RAG Architectures
Backend Frameworks	Vert.x, Node.js, Odoo, FastAPI
Distributed Systems	Kafka, Google Pub/Sub, Redis, HBase, MongoDB
Cloud & DevOps	AWS (Bedrock), Docker, Kubernetes, Git, CI/CD
Monitoring	Grafana, Prometheus
Soft Skills	Analytical Thinking, Problem Solving, Communication, Teamwork

Languages

English IELTS 6.5 (Dec 2024)

Education

2020 - 2024 **University of Science - Ho Chi Minh National University, Vietnam**
Bachelor of Information Technology (**GPA**: 3.7/4)

Research Experience

Research Engineer – Remote [Demo] Dec 2025 – March 2026

- **Knowledge Access & Retrieval:** Under the supervision of Dr. Srikanth Thudumu and in collaboration with To64, designed and implemented a hybrid graph+vector RAG system integrating Neo4j and Milvus with multimodal evidence retrieval.
- Built cross-modal retrieval and hybrid search flows for scientific queries, improving retrieval relevance, grounding, and contextual reasoning.

Collaborator – Computer Science Laboratory, HCMUS

- **Undergraduate Dissertation (Apr 2023 – Sep 2024)**: Supervised by Dr. Thanh Le; conducted a comprehensive survey on Temporal Knowledge Graphs (TKGs) and developed a deep learning-based approach for temporal reasoning.
- **Teaching Assistant – Part-time (Dec 2025 - Jan 2026)**: Assisted in grading and student support for undergraduate computer science courses.

Publications

Hoa Dao Tri Nguyen Phan, Thanh Le (2024). "Graph Convolution Transformer for Extrapolated Reasoning on Temporal Knowledge Graphs". **in** *CoopIS 2024*: URL: `https://www.insticc.org/node/TechnicalProgram/CoopIS/2024/presentationDetails/130018`.

Hoa Dao Tri Nguyen Phan, Thanh Le **and** Ngoc Trung Nguyen (2025). "HGCT: Enhancing Temporal Knowledge Graph Reasoning Through Extrapolated Historical Fact Extraction". **in** *Knowledge-Based Systems*: URL: `https://www.sciencedirect.com/science/article/pii/S0950705125004058`.

Work Experience

Software Engineer – Full-time, Momo M-service Aug 2025 – Present

- **OneLink System**: Worked in a team of three to maintain and enhance a deep-link platform serving ad, referral, and marketing traffic; improved reliability, monitoring, and debugging workflows.
- **Promotion Platform**: Supported backend service migration and system integration for user-acquisition campaigns, enabling seamless rollout and scalable campaign execution.

Software Developer – Freelance, BESCO Consulting Company September 2024 – March 2026

- **GKitchen**: Integrated ERP systems with **GrabMart**, **Grab Delivery**, and **Haravan**, enabling seamless order synchronization and operational automation.
- **PGI**: Worked in a team of two to design and develop ERP promotion and campaign features that supported and optimized sales operations.
- **Viet Thuong Music**: Worked in a team of five to integrate ERP systems with **Haravan** (e-commerce platform), **Sepay** (payment gateway), and **ZNS** (for OTP messaging and transaction notifications).

Collaborator – Robotics & IoT Club, HCMUS March 2023 – 2024

- Supported club activities, contributing to community engagement in robotics and IoT initiatives.

Translator – IELTS Song Ngu 2019 – 2022

- Translated English books, papers, and videos into Vietnamese, including Cambridge IELTS materials, The Economist articles, and TED-Ed videos.

Activities & Workshops

Attend AWS GameDay: Generative AI Unicorn Party April 2026

- Jointly organized by Momo, AWS, and G-AsiaPacific.
- Collaborated in a team of four to build and optimize AI-driven solutions using Amazon Bedrock, Amazon S3, and AWS Lambda in a gamified, real-time production environment.
- Solved complex technical quests involving multimodal RAG, agentic workflows, and cost-optimized infrastructure scaling.

Attend workshop on Industrial Internet of Things March 2024

- Jointly organized by Kyoto Institute of Technology and University of Science, HCMC, focusing on practical IIoT applications.
- Collaborated with four peers to develop a temperature monitoring application that alerts users when device temperatures exceed predefined limits.

Achievements

2020–2023	**Semester Scholarship**	IT Faculty, University of Science
2022–2023	**DXC Vietnam Scholarship**	DXC Vietnam
2022–2023	**Top 20 – Fem-lead Activator Scholarship**	Bosch Global Software Technologies Vietnam
2025	**MoMo Talent Program Graduate**, Backend Engineering Track	MoMo (M_Service)
2026	**Top 10 Finalist**, AWS GameDay: Generative AI Unicorn Party	AWS & MoMo & G-AsiaPacific